1hr

Roll Number:.....

National Institute of Technology Kurukshetra
End-Semester Examination, December 2024

Month and year: **Nov/Dec 2024**
Program: **B.Tech.**
Subject: **Problems Solving and Programming Using C**
Maximum Marks: **50**
Number of Questions to be attempted: **5**

Total no. of pages used: **2**
Semester: **1st**
Course code: **CSIC 103**
Timeallowed: **3 Hrs**
Total No of Questions: **5**

Instructions: All questions are Compulsory.
Attempts all parts of a question together.

1.	(a)	Write a C program that calculates mileage reimbursement for a salesperson at a rate of Rs. 16 per kilometer. *Sample Output: Your program should interact with the user in this manner:* *MILEAGE REIMBURSEMENT CALCULATOR* *Enter beginning odometer reading=> 13505.2* *Enter ending odometer reading=> 13810.6* *You traveled 305.4 kilometers. At Rs. 16 per km, your reimbursement is Rs 4886.40.*	[5]
	(b)	In a company, an employee is paid as follows: If his/her basic salary is less than Rs. 100000, then HRA is 10% of the base salary. If the salary is either equal to or above Rs. 100000, then HRA is Rs. 10000of the basic salary. The DA is fixed at 53%. If the Employee's salary is input, write a C program to find his Gross Salary. [Hint: Gross Salary = Basic Salary + HRA + DA]	[5]
2.	(a)	Any year is entered through the keyboard. Write a C function to determine whether the year is a leap year or not. **OR** What are the various mode of opening a file in C? Explain with examples.	[5]
	(b)	Given **N** as input from the user, write a C program to find the sum of all odd numbers that lie between 1 and **N.** [Hint: if N=5, the sum of odd numbers from 1 to 5 is 9 (1+3+5 = 9)]	[5]
3.	(a)	Write a C program to print the following pattern given the number of rows as input from the user. *Sample Test Case:* Enter numbers of rows = 5 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1	[5]
	(b)	Write a C function that finds the largest and the smallest elements in a N x M matrix, where N, M entered by the user.	[5]
4.	(a)	Design a function split, which takes a string and a character as argument and splits the string based on the character into multiple smaller size strings. The character should not be part of any of the split strings. Write a C program that calls the split function. *Sample Test Cases:* spilt("I am inside the class", ' '): should split the string to "I", "am", "inside", "the", "class" split("Walk faster. We are late", '.'): Should split the string to "Walk faster", "We are late"	[5]

	(b)	Create a structure named **Book** to store book details like *title*, *author*, and *price*. Write a C program to input details for four books, find the most expensive and the lowest priced books, and display their information.	[5]
5.	(a)	Rajiv is managing attendance for a student. The student either attends or misses a class. He logs the attendance of the student in anarray ***attendance[]*** of size 30 where 1 indicates the student was present and 0 indicates the student was absent. Rajiv wants to check whether the student is absent for more than or equal to **K** consecutive classes. If yes, Rajiv will send a warning to that studentwith the position (index) of the first missed class in the sequence.Write a C program to identify to check whether the student has more than or equal to**K** consecutive absences and print a warning for him/her. *Sample Test Case:* *Input: attendance[] = [1, 1, 0, 0, 0, 1, 1, 0]K = 3* *Output: You have more than or equal to3 consecutive absences starting from index 2.*	[5]
	(b)	What will be the output of the following programs?	[5]

i)

```c
#include<stdio.h>
void main()
{ int x=10, y=7, z, z1;
  z = (x! =4) || (y++==2);
  z1=(x++! =4) || (y++==2);
  printf("z=%d,z1=%d, s=%d",z,z1,x+y);
}
```

ii)

```c
#include<stdio.h>
#define sq(x) x*x
void main()
{
  printf("%d,%d",sq(2+2),64/sq(2+2));
}
```

iii)

```c
#include<stdio.h>
void main()
{
  for(int i=0;i<=25;i++)
   {
     switch(i++)
      {
        case 1:
        i+=2;
        case 2:
        i+=3;
        default:
        i+=4;
        printf("%d\n",i);
      }
   }
}
```

iv)

```c
#include<stdio.h>
void fun(int*x, int*y, int*z)
{
  *x=*x+10;
  x=y;
  *x=*x+*y+*z;
  *z=*x+*y+*z;
}
void main()
{
  int x=10,y=20;
  fun(&x, &y, &x);
  printf("%d",x+y);
}
```

v)

```c
#include<stdio.h>
struct node
{
  int nid;
  char nname[10];
};
void main()
{
  struct node n;
  printf("%ld,%ld",sizeof(n),sizeof(n.nname));
}
```